

Questions with Answer Keys

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Q1 - Single Correct

If $\frac{\sin \theta}{\cos 3\theta} + \frac{\sin 3\theta}{\cos 9\theta} + \frac{\sin 9\theta}{\cos 27\theta} = \frac{1}{2}(\tan x - \tan \theta)$ then the value of x is

- (1) 3θ
- (2) 6θ
- (3) 9θ
- (4) 27θ

Q2 - Single Correct

Minimum value of $4x^2 - 4x|\sin \theta| - \cos^2 \theta$ is equal to

- (1) -2
- (2) -1
- (3) -1/2
- (4) 0

Q3 - Single Correct

The value of $\cos^4\left(\frac{\pi}{8}\right) + \cos^4\left(\frac{3\pi}{8}\right) + \cos^4\left(\frac{5\pi}{8}\right) + \cos^4\left(\frac{7\pi}{8}\right)$ is equal to

- (1) 1/2
- (2) 3/4
- (3) 3/2
- (4) 1

Q4 - Single Correct

$\sin 47^\circ + \sin 61^\circ - \sin 11^\circ - \sin 25^\circ$ is equal to

- (1) $\sin 36^\circ$
- (2) $\cos 36^\circ$
- (3) $\sin 7^\circ$
- (4) $\cos 7^\circ$

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Q5 - Single Correct

If $\tan \theta_1, \tan \theta_2, \tan \theta_3$ and $\tan \theta_4$ are the roots of the equation

$$x^4 - x^3 \sin 2\beta + x^2 \cos 2\beta - x \cos \beta - \sin \beta = 0 \text{ then } \tan(\theta_1 + \theta_2 + \theta_3 + \theta_4) \text{ is equal to}$$

(1) $\sin \beta$

(2) $\cos \beta$

(3) $\tan \beta$

(4) $\cot \beta$

Q6 - Single Correct

The value of $\sum_{k=1}^{13} \frac{1}{\sin\left(\frac{\pi}{4} + \frac{(k-1)\pi}{6}\right) \sin\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)}$ is equal to

(1) $3 - \sqrt{3}$

(2) $2(3 - \sqrt{3})$

(3) $2(\sqrt{3} - 1)$

(4) $2(2 + \sqrt{3})$

Q7 - Single Correct

The maximum value of $(\cos \alpha_1)(\cos \alpha_2) \cdots (\cos \alpha_n)$ under the restriction $0 \leq \alpha_1, \alpha_2, \dots, \alpha_n \leq \pi/2$ and $(\cot \alpha_1)(\cot \alpha_2) \cdots (\cot \alpha_n) = 1$ is

(1) $\frac{1}{2^{n/2}}$

(2) $\frac{1}{2^n}$

(3) $\frac{1}{2n}$

(4) 1

Q8 - Single Correct

If $x_{n+1} = \sqrt{\frac{1}{2}(1 + x_n)}$, then $\cos \left[\frac{\sqrt{1-x_0^2}}{x_1 x_2 x_3 \dots \text{to infinite}} \right] (-1 < x_0 < 1)$ is equal to

(1) -1

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(2) 1

(3) x_0

(4) $1/x_0$

Q9 - Single Correct

Let $S = \left\{ x \in (-\pi, \pi) : x \neq 0, \pm \frac{\pi}{2} \right\}$, the sum of all distinct solutions of the equation

$\sqrt{3} \sec x + \operatorname{cosec} x + 2(\tan x - \cot x) = 0$ in the set S is equal to

(1) $-\frac{7\pi}{9}$

(2) $-\frac{2\pi}{9}$

(3) 0

(4) $\frac{5\pi}{9}$

Q10 - Single Correct

For $x \in (0, \pi)$, the equation $\sin x + 2 \sin 2x - \sin 3x = 3$ has

(1) infinitely many solutions

(2) three solutions

(3) one solution

(4) no solution

Q11 - Single Correct

The number of solutions of the pair of equations $2 \sin^2 \theta - \cos 2\theta = 0$ and $2 \cos^2 \theta - 3 \sin \theta = 0$ in the

interval $[0, 2\pi]$ is

(1) 0

(2) 1

(3) 2

(4) 4

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Q12 - Single Correct

Number of solutions of the equation $\cos(\pi\sqrt{x}) \cdot \cos(\pi\sqrt{x-4}) = 1$ are

- (1) 1
- (2) 2
- (3) 3
- (4) 0

Q13 - Single Correct

If $\left| 1 - \frac{|\sin x|}{|1 + \sin x|} \right| \geq \frac{2}{3}$, then $\sin x$ lies in

- (1) $\left(-\infty, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$
- (2) $\left(-\frac{1}{2}, \frac{1}{2}\right)$
- (3) $\left[-\frac{1}{2}, \frac{1}{2}\right]$
- (4) none of these

Q14 - Single Correct

The value of $\cos^{-1} x + \cos^{-1} \left(\frac{x}{2} + \frac{1}{2} \sqrt{3 - 3x^2} \right)$ ($1/2 \leq x \leq 1$) is equal to

- (1) $\pi/6$
- (2) $\pi/3$
- (3) π
- (4) 0

Q15 - Single Correct

The sum of the infinite series $\cot^{-1} 2 + \cot^{-1} 8 + \cot^{-1} 18 + \cot^{-1} 32 + \dots$ is equal to

- (1) $\pi/3$
- (2) $\pi/4$
- (3) $\pi/6$

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(4) $\pi/8$

Q16 - Single Correct

If $a \sin^{-1} x - b \cos^{-1} x = c$, then $a \sin^{-1} x + b \cos^{-1} x$ is equal to

(1) 0

(2) $\frac{\pi ab + c(b-a)}{a+b}$

(3) $\pi/2$

(4) $\frac{\pi ab + c(a-b)}{a+b}$

Q17 - Single Correct

The equation $2 \cos^{-1} x = \sin^{-1} (2x\sqrt{1-x^2})$ is valid for all values of x satisfying

(1) $-1 \leq x \leq 1$

(2) $0 \leq x \leq 1$

(3) $0 \leq x \leq 1/\sqrt{2}$

(4) $1/\sqrt{2} \leq x \leq 1$

Q18 - Multiple Correct

If $A = \tan^{-1}(1/7)$ and $B = \tan^{-1}(1/3)$, then

(1) $\cos 2A = 24/25$

(2) $\cos 2B = 4/5$

(3) $\cos 2A = \sin 4B$

(4) $\tan 2B = 3/4$

Q19 - Multiple Correct

$\sin^{-1} 6x + \sin^{-1} 6\sqrt{3}x = -\pi/2$ if x is equal to

(1) $-1/12$

(2) $1/6$

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(3) $1/12$

(4) $-1/6$

Q20 - Multiple Correct

$\theta = \tan^{-1}(2 \tan^2 \theta) - \tan^{-1}((1/3) \tan \theta)$ if $\tan \theta$ is equal to

(1) -2

(2) 1

(3) $2/3$

(4) 2

Q21 - Multiple Correct

If x_1, x_2, x_3, x_4 are roots of the equation $x^4 - x^3 \sin 2\beta + x^2 \cos 2\beta - x \cos \beta - \sin \beta = 0$ then

$\sum_{i=1}^4 \tan^{-1} x_i$ is equal to

(1) $\pi - \beta$

(2) $\pi + 2\beta$

(3) $(\pi/2) - \beta$

(4) $(\pi/2) - 2\beta$

Q22 - Multiple Correct

A solution of the equation $(1 - \tan \theta)(1 + \tan \theta) \sec^2 \theta + 2^{\tan 2\theta} = 0$ where θ lies in the interval $(-\pi/2, \pi/2)$ is given by.

(1) $\theta = 0$

(2) $\theta = \pi/3$

(3) $\theta = -\pi/3$

(4) $\theta = \pi/6$

Q23 - Multiple Correct

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The equation $81^{\sin^2 x} + 81^{\cos^2 x} = 30$ has two roots in the

- (1) I quadrant
- (2) II quadrant
- (3) III quadrant
- (4) IV quadrant

Q24 - Multiple Correct

Let $\theta, \phi \in [0, 2\pi]$ be such that $2 \cos \theta (1 - \sin \phi) = \sin^2 \theta \left(\tan \frac{\theta}{2} + \cot \frac{\theta}{2} \right) \cos \phi - 1$ and $\tan(2\pi - \theta) > 0$ and $-1 < \sin \theta < -\frac{\sqrt{3}}{2}$, then ϕ cannot satisfy

- (1) $0 < \phi < \frac{\pi}{2}$
- (2) $\frac{\pi}{2} < \phi < \frac{4\pi}{3}$
- (3) $\frac{4\pi}{3} < \phi < \frac{3\pi}{2}$
- (4) $\frac{3\pi}{2} < \phi < 2\pi$

Q25 - Multiple Correct

If $\tan x = 2b/(a - c)$ ($a \neq c$), $y = a \cos^2 x + 2b \sin x \cos x + c \sin^2 x$ and $z = a \sin^2 x - 2b \sin x \cos x + c \cos^2 x$, then

- (1) $y = z$
- (2) $y + z = a + c$
- (3) $y - z = a - c$
- (4) $y - z = (a - c)^2 + 4b^2$.

Q26 - Multiple Correct

If $x = \sqrt{a^2 \cos^2 \alpha + b^2 \sin^2 \alpha} + \sqrt{a^2 \sin^2 \alpha + b^2 \cos^2 \alpha}$ then $x^2 = a^2 + b^2 + 2\sqrt{P(a^2 + b^2) - P^2}$ where P is equal to

- (1) $a^2 \cos^2 \alpha + b^2 \sin^2 \alpha$
- (2) $a^2 \sin^2 \alpha + b^2 \cos^2 \alpha$.

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(3) $(1/2) [a^2 + b^2 + (a^2 - b^2) \cos 2\alpha]$

(4) $(1/2) [a^2 + b^2 - (a^2 - b^2) \cos 2\alpha]$

Q27 - Multiple Correct

If $\cos(\beta - \gamma) + \cos(\gamma - \alpha) + \cos(\alpha - \beta) = -3/2$ then

(1) $\Sigma \cos \alpha = 0$

(2) $\Sigma \sin \alpha = 0$

(3) $\Sigma \cos \alpha \sin \alpha = 0$

(4) $\Sigma(\cos \alpha + \sin \alpha) = 0$

Q28 - Paragraph 1

Passage I (For Question 28, 29)

$\cos \frac{\pi}{7}, \cos \frac{3\pi}{7}, \cos \frac{5\pi}{7}$ are the roots of the equation $8x^3 - 4x^2 - 4x + 1 = 0$

The value of $\sec \frac{\pi}{7} + \sec \frac{3\pi}{7} + \sec \frac{5\pi}{7}$ is

(1) 2

(2) 4

(3) 8

(4) none of these

Q29 - Paragraph 1

The value of $\sin \frac{\pi}{14} \sin \frac{3\pi}{14} \sin \frac{5\pi}{14}$ is

(1) $\frac{1}{4}$

(2) $\frac{1}{8}$

(3) $\frac{\sqrt{7}}{4}$

(4) $\frac{\sqrt{7}}{8}$

Q30 - Paragraph 2

Passage II (For Question 30, 31)

ABC is a triangle, O is a point inside the triangle so that its distance from A, B, C is respectively a, b, c . L, M, N are the feet of the perpendiculars from O to AB, BC, CA respectively. x, y, z are respectively the distances of O from L, M, N

$$\angle OAL = \alpha, \angle OBM = \beta, \angle OCN = \gamma$$

$\sin(A - \alpha) + \sin(B - \beta) + \sin(C - \gamma)$ is equal to

(1) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c}$

(2) $\frac{x}{b} + \frac{y}{c} + \frac{z}{a}$

(3) $\frac{x}{c} + \frac{y}{a} + \frac{z}{b}$

(4) none of these

Q31 - Paragraph 2

$a + b + c$ is equal to

(1) $x \sin \alpha + y \sin \beta + z \sin \gamma$

(2) $x \cos \alpha + y \cos \beta + z \cos \gamma$

(3) $x \operatorname{cosec} \alpha + y \operatorname{cosec} \beta + z \operatorname{cosec} \gamma$

(4) $x \sec \alpha + y \sec \beta + z \sec \gamma$

Q32 - Paragraph 3

Passage III (For Question 32, 33)

$$f(x) = \sin\{\cot^{-1}(x+1)\} - \cos(\tan^{-1}x)$$

$$a = \cos \tan^{-1} \sin \cot^{-1} x$$

$$b = \cos(2 \cos^{-1} x + \sin^{-1} x)$$

The value of x for which $f(x) = 0$ is

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(1) $-1/2$

(2) 0

(3) $1/2$

(4) 1

Q33 - Paragraph 3

If $f(x) = 0$ then a^2 is equal to

(1) $1/2$

(2) $2/3$

(3) $5/9$

(4) $9/5$

Q34 - Paragraph 4

Passage IV (For Question 34, 35)

α is a root of the equation

$$(2 \sin x - \cos x)(1 + \cos x) = \sin^2 x$$

β is a root of the equation $3 \cos^2 x - 10 \cos x + 3 = 0$ and γ is a root of the equation

$$1 - \sin 2x = \cos x - \sin x, 0 \leq \alpha, \beta, \gamma \leq \pi/2$$

$\sin \alpha + \sin \beta + \sin \gamma$ can be equal to

(1) $\frac{14+3\sqrt{2}}{6\sqrt{2}}$

(2) $5/6$

(3) $\frac{3+4\sqrt{2}}{6}$

(4) $\frac{1+\sqrt{2}}{2}$

Q35 - Paragraph 4

$\sin(\alpha - \beta)$ is equal to

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(1) 1

(2) 0

(3) $\frac{1-2\sqrt{6}}{6}$

(4) $\frac{\sqrt{3}-2\sqrt{2}}{6}$

Q36 - Paragraph 5

Passage V (For Question 36, 37)

$$\cos \theta \cos 2\theta \cos 3\theta = 1/4 (0 \leq \theta \leq \pi)$$

Sum of the roots of this equation is

(1) π

(2) 2π

(3) 3π

(4) 4π

Q37 - Paragraph 5

If α is a root of this equation, $2 \cos \alpha$ is a root of the equation

(1) $x^2 - 1 = 0$

(2) $x^2 + 1 = 0$

(3) $x^4 - 4x^2 + 2 = 0$

(4) $x^4 - 4x^2 + 3 = 0$

Q38 - Integer Type

The number of distinct solutions of the equation $\frac{5}{4} \cos^2 2x + \cos^4 x + \sin^4 x + \cos^6 x + \sin^6 x = 2$ in the interval $[0, 2\pi]$ is

Q39 - Integer Type

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The number of value of θ in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ such that $\theta \neq \frac{n\pi}{5}$ for $n = 0, \pm 1, \pm 2$ and $\tan \theta = \cot 5\theta$ as well as $\sin 2\theta = \cos 4\theta$ is.

Q40 - Integer Type

The number of all possible values of θ , where $0 < \theta < \pi$, for which the system of equations

$$(y+z)\cos 3\theta = (xyz)\sin 3\theta$$

$$x\sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z}$$

$$(xyz)\sin 3\theta = (y+2z)\cos 3\theta + y\sin 3\theta$$

Have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$

Q41 - Integer Type

Let $f : [0, 4\pi] \rightarrow [0, \pi]$ be defined by $f(x) = \cos^{-1}(\cos x)$. The number of points $x \in [0, 4\pi]$ satisfying the equation $f(x) = \frac{10-x}{10}$ is

Q42 - Integer Type

If $x = \sec^2(\tan^{-1} 2) + \operatorname{cosec}^2(\cot^{-1} 3)$ and $y = \tan^{-1} \frac{\sqrt{1+\tan^2 8}-1}{\tan 8}$ then $15x^2 + 4y^2$ is equal to $3434 + \lambda$ where $\lambda =$

Q43 - Integer Type

If $\cos \theta + \cos^2 \theta + \cos^3 \theta = 1$ and $\sin^6 \theta = a + b\sin^2 \theta + c\sin^4 \theta$, then $a + b + c =$

Q44 - Integer Type

If $K = \sin^3 10^\circ + \sin^3 50^\circ - \sin^3 70^\circ$, then $8K + 6 =$

Q45 - Integer Type

If $A + B + C = 180^\circ$, $\frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C} = k \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$ then the value of $k^3 - 8k^2 + 1$ is equal to

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Answer Key

Q1 (4)

Q2 (2)

Q3 (3)

Q4 (4)

Q5 (4)

Q6 (3)

Q7 (1)

Q8 (3)

Q9 (3)

Q10 (4)

Q11 (3)

Q12 (1)

Q13 (3)

Q14 (2)

Q15 (2)

Q16 (4)

Q17 (4)

Q18 (1,2,3,4)

Q19 (1)

Q20 (1,2)

Q21 (3)

Q22 (2,3)

Q23 (1,2,3,4)

Q24 (1,3,4)

Q25 (2,3)

Q26 (1,2,3,4)

Q27 (1,2,4)

Q28 (2)

Q29 (2)

Q30 (2)

Q31 (3)

Q32 (1)

Q33 (3)

Q34 (1,3)

Q35 (3)

Q36 (3)

Q37 (1,3)

Q38 (8)

Q39 (3)

Q40 (3)

Q41 (3)

Q42 (5)

Q43 (0)

Q44 (3)

Q45 (1)